

Nexidia Real-Time Grid Silent Installer Guide

Version 12.5.1

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Running Silent Installer

Using the batch file

In the folder that contains the installation files, there is a **SilentInstallRTG.bat** that will run the silent installer with the following parameters

- `--srtFile [filename]`, Optional parameter where filename is the Site Readiness Tool JSON file to use, the default JSON file is **silent_install_properties.json** in the directory with the installation files
- `--listServers`, Optional parameter that will list the servers in the JSON file

The Silent Installer will uninstall the current Real-Time Grid services and install the services passed into the Silent Installer by the identity parameter

Using Powershell Remotely

1. Copy the installation folder with the Keystore, Truststore, and SRT Properties JSON files to the remote computer
2. Open Powershell
3. Run the command:

```
Invoke-Command -ComputerName [remote_computer_name] -ScriptBlock { cmd /c [silent_install_batch_file] } SHELL SCRIPT
```

- **remote_computer_name** - the fully qualified domain name of the computer to run the Silent Installer on
- **silent_install_batch_file** - the absolute file name of the SilentInstallRTG.bat double-quoted

NOTE

The user that you are logged in as will be the user that runs the installer on the remote computer. If you do not have Administrator privileges, you can change the user by adding `-Credential [user_with_domain]`.

4. The Silent Installer will return **Install Finished Successfully** if the installation is successful or **Install Failed** if any problems occur during the installation

Site Readiness Tool

A file named **silent_install_properties.json** with all the properties needed to set up each of the Real-Time Grid services

- Message Broker
- Gateway
- Config

- Telephony Proxy
- Scan

Real-Time Grid Properties

Common Properties

Property Name	Description
id	GUID representing the Real-Time Grid service
hostname	The host where the node will be installed on. Used to determine if the service is installed. The host name can be only the name of the server or the fully qualified domain name.
serviceAccountUser	The Domain User that will run the Windows service. Required for all Real-Time Grid nodes.
serviceAccountPwd	The Domain User's password that runs the Windows service. Required for all Real-Time Grid nodes.
installDir	Container configuration for the service

Installer Configuration Properties

Property Name	Description
telephonyEnvironment	The telephony environment RTG will connect to, valid values are Engage or Mattersight
installDir	Directory to install RTG. The default directory is C:\Program Files\Nexidia\Real-Time Grid
dataFolder	The RTG data directory for logs, workflows, configurations, and agent mappings. The default directory is C:\ProgramData\Nexidia\Real-Time Grid
encryptionDataFolder	The directory on an encrypted drive to store logs, agent mappings, and append metadata files. The default directory is C:\ProgramData\Nexidia\Real-Time Grid
messageBrokerUserCredentials	
userName	The username to connect to the message broker
password	The encrypted password to connect to the message broker
generalSettings	General Settings, Config, Telephony Proxy and Scan.
defaultIP	The IP Address hosting the node
messageBrokerIPAddress	The IP Address hosting the message broker

Property Name	Description
messageBrokerPort	The port of the message broker
gatewayServiceIpAddress	The IP address hosting the Gateway service
gatewayServicePort	The port of the Gateway service
configServiceIpAddress	The IP address hosting the Config service
configServicePort	The port used by the Config service
securitySettings	
securityEnabled	Enables the ability to activate all Real Time Grid security features (HTTPS and SIPS) when set to true. Disables all security features when set to false.
httpsEnabled	When enabled true RTG utilizes TLS/SSL for all HTTP layer communication (HTTPS).
messageBrokerTlsEnabled	When enabled true RTG utilizes TLS to connect to the message broker
sipsEnabled	When enabled true RTG utilizes TLS/SSL for all SIP layer communication (SIPS).
loginMethod	Method used to log into Mattersight or Engage ConnectAPI. Possible values are PasswordAndDomainLogin or SSOLogin . For Engage environments use PasswordAndDomainLogin for ClientLoginWithDomain, or SSOLogin for ClientLoginSSO.
domain	Name of the current RTG service user's domain. Required if Scan service logs into Engage ConnectAPI service using ClientLoginWithDomain method
keyStoreFile	Name of the keystore file used for encrypting passwords and TLS/SSL communication. This file should be in the directory where the installation is run from.
keyStoreType	KeyStore type used for TLS/SSL communication. PKCS12 or JCEKS types are supported.
keyStorePassword	The encrypted keystore password generated by the Password Generator to the Java keystore file used for TLS/SSL communication. Use the output result when the parameter showEncryptedPassphrase is used with the Password Generator
trustStoreFile	Name of the truststore file. This file should be in the directory where the installation is run from.
trustStoreType	TrustStore type used for TLS/SSL communication. PKCS12 , JCEKS or JKS types are supported.
trustStorePassword	The encrypted password generated by the Password Generator to the Java truststore file used for TLS/SSL communication.
engageScan	Required for Scan and Telephony Proxy when telephonyEnvironment set to Engage
connectApiHost	The IP address of the server hosting ConnectAPI. Required for Scan and Telephony Proxy

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Property Name	Description
connectAPIUser	The user who logs into ConnectAPI. Required for Scan and Telephony Proxy
connectAPIPassword	The ConnectAPI password of the ConnectAPI user. Required for Scan and Telephony Proxy
agentType	The agent type used by the customer, valid values are Agent or Extension . Required for Telephony Proxy
mattersightEnvironment	Required for Scan and Telephony Proxy when telephonyEnvironment set to Mattersight
mattersightHost	The FQDN of the server hosting Mattersight. Required for Scan and Telephony Proxy
mattersightPort	The listening port of the Mattersight server. Required for Scan and Telephony Proxy
mattersightUser	The username logs into the Mattersight server. Required for Scan and Telephony Proxy
mattersightPassword	The password of the Mattersight user. Required for Scan and Telephony Proxy
receiveHost	The FQDN of the server that receives messages from Mattersight. Required for Telephony Proxy
receivePort	The listening port of the server that receives messages from Mattersight. Required for Telephony Proxy
acdlids	A comma delimited string indicating which acds are present and will be used to receive call events from Mattersight. Required for Telephony Proxy
scanSettings	
scanServerFQDN	The Fully Qualified Domain Name of the scan server.
scanActiveCalls	The active calls the scan server can allow.
publishEnlightenScores	Whether scan will publish Enlighten scores.
publishEnlightenScoresFrequency	Enlighten score updates, in seconds.
publishTranscripts	Whether scan will publish Transcripts.
publishTranscriptsFrequency	Transcripts, in seconds.
publishTranscriptsFinalizing	Whether scan will publish finalized Transcripts.
publishTranscriptsFinalizingFrequency	Finalized Transcripts, in seconds.
sipPort	The port scan uses to send SIP requests to Engage AIR loggers.
sipPort2	The port scan uses to receive SIP responses from Engage AIR loggers or Mattersight.
messageBrokerAddress	Required for Message Broker
messageBrokerAddress	The message broker server that other services will connect to

Property Name	Description
<code>globalMaxSize</code>	Memory allocated to the message broker; use Mb for Megabytes and Gb for Gigabytes. Only whole numbers are allowed, no decimals
<code>messageBrokerConnections</code>	Message broker connections that other services will connect to. This is a section in the <code>messageBrokerSettings</code> section
<code>name</code>	The unique name of the connection
<code>port</code>	The unique listening port of the connection
<code>protocol</code>	The protocol of the connection. Valid values are NIO , AMQP , or STOMP
<code>tlsEnabled</code>	When set to true, the connection will only allow services that have TLS/SSL communication set up. Requires keystore and truststore files defined in the <code>storeSettings</code> section
<code>storeSettings</code>	The settings section in the <code>messageBrokerSettings</code> section
<code>keyStoreFile</code>	Name of the keystore file used for encrypting passwords and TLS/SSL communication. This file should be in the directory where the installation is run from.
<code>keyStoreType</code>	KeyStore type used for TLS/SSL communication. PKCS12 or JCEKS types are supported.
<code>keyStorePassword</code>	The encrypted keystore password generated by the Password Generator to the Java keystore file used for TLS/SSL communication. Use the output result when the parameter showEncryptedPassphrase is used with the Password Generator
<code>trustStoreFile</code>	Name of the truststore file. This file should be in the directory where the installation is run from.
<code>trustStoreType</code>	TrustStore type used for TLS/SSL communication. PKCS12 , JCEKS or JKS types are supported.
<code>trustStorePassword</code>	The encrypted password generated by the Password Generator to the Java truststore file used for TLS/SSL communication.

Example `silent_install_properties.json`


```

{
  "scanNodes": [
    {
      "id": "7C9B209E-E92B-4A28-98EB-34B012BD7313",
      "hostname": "scan1.ex-node.com",
      "serviceAcctUser": "test-domain\\winsuser",
      "serviceAcctPwd": "Q0uguZAxquiZ8Wug==",
      "installerConfiguration" : {
        "telephonyEnvironment" : "engage",
        "generalSettings" : {
          "defaultIpAddress" : "10.1.1.55",
          "messageBrokerIpAddress" : "10.1.1.58",
          "messageBrokerPort" : "26011",
          "configServiceIpAddress" : "10.1.1.57",
          "configServicePort" : "26002",
          "gatewayServiceIpAddress" : "10.1.1.57",
          "gatewayServicePort" : "26022"
        },
        "securitySettings" : {
          "securityEnabled" : true,
          "loginMethod" : "PasswordAndDomainLogin",
          "httpsEnabled" : true,
          "messageBrokerTlsEnabled" : true,
          "trustStoreFile" : "cacerts",
          "trustStoreType" : "JKS",
          "trustStorePassword" : "Q0uguZAxquiZ8Wug==",
          "keyStoreFile" : "scan.jceks.keystore",
          "keyStoreType" : "JCEKS",
          "keyStorePassword" : "h6rcHge5XLFGDwJD6Q==",
          "domain" : "FAST-TALK",
          "sipsEnabled" : true
        },
        "messageBrokerUserCredentials" : {
          "username" : "admin",
          "password" : "ltrHsdRP7+gdGnTDg=="
        },
        "engageSettings" : {
          "connectApiHost" : "capi.server.com",
          "connectApiUser" : "capiUser",
          "connectApiPassword" : "Q0uguZAxquiZ8Wug=="
        },
        "scanSettings" : {
          "fqdn" : "scan1.ex-node.com",
          "maxActiveCalls": 100,
          "publishEnlightenScores" : true,
          "publishEnlightenScoresFreqSecs" : 10,
          "publishTranscripts" : true,
          "publishTranscriptsFreqSecs": 15,
          "publishTranscriptsFinalizing" : true,
          "publishTranscriptsFinalizingFreqSecs": 20,
          "streamingMode": "LOW_LATENCY",
          "sipUasPort" : "5525",
          "sipUacPort" : "6652"
        }
      }
    },
    {
      "id": 2,
      "hostname": "scan2.ex-node.com",
      "serviceAcctUser": "test-domain\\winsuser",
      "serviceAcctPwd": "Q0uguZAxquiZ8Wug==",
      "installerConfiguration" : {

```

```

    "telephonyEnvironment" : "engage",
    "generalSettings" : {
"defaultIpAddress" : "10.1.1.56",
"messageBrokerIpAddress" : "10.1.1.58",
"messageBrokerPort" : "26011",
"configServiceIpAddress" : "10.1.1.57",
"configServicePort" : "26002",
"gatewayServiceIpAddress" : "10.1.1.57",
"gatewayServicePort" : "26022"
    },
    "securitySettings" : {
"securityEnabled" : true,
"loginMethod" : "PasswordAndDomainLogin",
"httpsEnabled" : true,
"messageBrokerTlsEnabled" : true,
"trustStoreFile" : "cacerts",
"trustStoreType" : "JKS",
"trustStorePassword" : "Q0uguZAxquiZ8Wug==",
"keyStoreFile" : "scan.jceks.keystore",
"keyStoreType" : "JCEKS",
"keyStorePassword" : "h6rcHge5XLFGDwJD6Q==",
"domain" : "FAST-TALK",
"sipsEnabled" : true
    },
    "messageBrokerUserCredentials" : {
"username" : "admin",
"password" : "ltrHsdRP7+gdGnTDg=="
    },
    "engageSettings" : {
"connectApiHost" : "capi.server.com",
"connectApiUser" : "capiUser",
"connectApiPassword" : "Q0uguZAxQafAkPquiZ8Wug=="
    },
    "scanSettings" : {
"fqdn" : "scan2.ex-node.com",
"maxActiveCalls": 100,
"publishEnlightenScores" : true,
"publishEnlightenScoresFreqSecs" : 10,
"publishTranscripts" : true,
"publishTranscriptsFreqSecs" : 15,
"publishTranscriptsFinalizing" : true,
"publishTranscriptsFinalizingFreqSecs": 20,
"streamingMode": "LOW_LATENCY",
"sipUasPort" : "1234",
"sipUacPort" : "6525"
    }
}
],
"gatewayNodes": [
{
    "id": "7FC04576-F2A7-4476-910D-4ABCD50176B9",
    "hostname": "rtg-services.ex-node.com",
    "serviceAcctUser": "test-domain\\winsuser",
    "serviceAcctPwd": "Q0uguZAxquiZ8Wug==",
    "installerConfiguration" : {
"telephonyEnvironment" : "engage",
"generalSettings" : {
"defaultIpAddress" : "10.1.1.57",
"messageBrokerIpAddress" : "10.1.1.58",
"messageBrokerPort" : "26011",
"configServiceIpAddress" : "10.1.1.57"

```

```

"configServicePort" : "26002",
"gatewayServiceIpAddress" : "10.1.1.57",
"gatewayServicePort" : "26022"
},
"messageBrokerUserCredentials" : {
  "username" : "admin",
  "password" : "ltrHsdRP7+gdGnTDg=="
},
"securitySettings" : {
"securityEnabled" : true,
"loginMethod" : "PasswordAndDomainLogin",
"httpsEnabled" : true,
"messageBrokerTlsEnabled" : true,
"trustStoreFile" : "cacerts",
"trustStoreType" : "JKS",
"trustStorePassword" : "Q0uguZAxquiZ8Wug==",
"keyStoreFile" : "scan.jceks.keystore",
"keyStoreType" : "JCEKS",
"keyStorePassword" : "h6rcHge5XLFGDwJD6Q==",
"domain" : "FAST-TALK",
"sipsEnabled" : true
}
}
},
"telephonyProxyNodes": [
{
  "id": 4,
  "hostname": "rtg-services.ex-node.com",
  "ports": "26011",
  "type": "primary",
  "serviceAcctUser": "test-domain\\winsuser",
  "serviceAcctPwd": "Q0uguZAxquiZ8Wug==",
  "installerConfiguration" : {
    "telephonyEnvironment" : "engage",
    "generalSettings" : {
      "defaultIpAddress" : "10.1.1.57",
      "messageBrokerIpAddress" : "10.1.1.58",
      "messageBrokerPort" : "26011",
      "configServiceIpAddress" : "10.1.1.57",
      "configServicePort" : "26002",
      "gatewayServiceIpAddress" : "10.1.1.57",
      "gatewayServicePort" : "26022"
    },
    "messageBrokerUserCredentials" : {
      "username" : "admin",
      "password" : "ltrHsdRP7+gdGnTDg=="
    },
    "securitySettings" : {
      "securityEnabled" : true,
      "loginMethod" : "PasswordAndDomainLogin",
      "httpsEnabled" : true,
      "messageBrokerTlsEnabled" : true,
      "trustStoreFile" : "cacerts",
      "trustStoreType" : "JKS",
      "trustStorePassword" : "Q0uguZAxquiZ8Wug==",
      "keyStoreFile" : "scan.jceks.keystore",
      "keyStoreType" : "JCEKS",
      "keyStorePassword" : "h6rcHge5XLFGDwJD6Q==",
      "domain" : "FAST-TALK",
      "sipsEnabled" : true
    }
  },

```

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```

"tlsEnabled" : false
}, {
"name" : "tls_conn",
"protocol" : "NIO",
"port" : "26011",
"tlsEnabled" : true
}],
"storeSettings" : {
"keyStoreFile" : "scan.jceks.keystore",
"keyStoreType" : "JCEKS",
"keyStorePassword" : "h6rcHge5XLFGDwJD6Q==",
"trustStoreFile" : "cacerts",
"trustStoreType" : "JKS",
"trustStorePassword" : "Q0uguZAxquiZ8Wug=="
}
}
}
}
]
}

```

Mattersight Example JSON

Replace the engageSettings sections with the example below

```

{
  "mattersightSettings" : {
    "mebHost" : "mattersight.server.local",
    "mebPort" : 7005,
    "mebUsername" : "admin",
    "mebPassword" : "Nwc9zeWLCpFLSAYVg",
    "receivingHost" : "10.1.1.57",
    "receivingPort" : 26015,
    "acdIds" : "ACD_0,ACD_1"
  }
}

```

JSON

Password Generator

InstallerDialogBannerImage

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A command line tool used to generate an encryption key in the keystore file and encrypt passwords stored in the properties files.

1. Navigate to the RTG installation bin directory. The default location is `C:\Program Files\Nexidia\Real-Time Grid 12.5\bin`. Open a command prompt in the directory.
2. Run the command `password-generator.cmd` with the following parameters:

```
-keystoreFilename <keystore_filename>
-keystorePassphrase <keystore_password>
-keystoreType <keystore_type>
-generateEncryptionKey
-showEncryptedPassphrase
-passwords <password_list>
```

`<keystore_filename>` is the path of the keystore file where the encryption key exists. The file must exist. Required

`<keystore_password>` is the password of the keystore file. Required

`<keystore_type>` is the keystore type. The choices are JCEKS (Default) or PKCS12. Optional

-generateEncryptionKey generates the encryption key and stores it in the file defined by the `<keystore_filename>` value in the **-keystoreFilename** parameter. Optional

-showEncryptedPassphrase outputs the keystore password defined by the `<keystore_password>` value in the **-keystorePassphrase** parameter. Optional

`<password_list>` comma delimited list of passwords or strings to encode. Optional

3. The output will show each password and its encrypted value.

The following example would encode a SQL connection string:

```
password-generator.cmd -keystoreFilename ./example.keystore -keystorePassphrase [unencrypted keystore password]
-keystoreType JCEKS -generateEncryptionKey -showEncryptedPassphrase -passwords
jdbc:sqlserver://example.sql.server.address:1433;user=example;password=examplepassword
```

Passphrase encrypted: [encrypted keystore password] Passwords Encrypted:

`jdbc:sqlserver://example.sql.server.address:1433;user=example;password=examplepassword`: [encrypted SQL connection string]

Keystore and Truststore Generation with Keystore Explorer

InstallerDialogBannerImage

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Pre-requisites

Software

- KeyStore Explorer

Keystore

- Certificate file for the Message Broker server
 - OpenSSL
 - PKCS#12
 - PKCS#8
- Private Key file, if using PKCS#8 or OpenSSL certificate

Truststore

- X.509 formatted Certificate Authority or Certificate for the hosting servers:
 - **Engage**
 - Message Broker server
 - ConnectAPI server
 - SQL server
 - AIR servers
 - **Mattersight**
 - Message Broker server
 - Mattersight Enlighten Bridge server

Creating a Keystore file

1. Open KeyStore Explorer
2. Click on File → New
 - Select JCEKS for the Keystore Type
3. Click on File → Save
 - a. Enter a password for the keystore file
 - b. Click OK
 - c. Enter the name of the Keystore file and click Save
4. Generate the encryption key

https://wem-dochub.mindtouch.us/Nexidia_Analytics_and_Data_Exchange_Framework/NXQC_12.5/Grid_Documentati...

Updated: Thu, 30 Apr 2026 18:15:27 GMT

- a. Click on Tools → Generate Secret Key
 - Set Algorithm to AES
 - Set Key Size to 128
 - Click OK
 - Set Alias to nexidia_encryption_key
 - Click OK
 - Enter the same password used for the Keystore itself
 - Click OK
5. Import the Message Broker certificate
 - To get the type of the certificate or key
 - a. Click on Examine → Detect File Type
 - For a certificate PFX file, select the certificate
 - For a certificate cer file, select the private key file, usually ending in .key
 - b. Click Detect Type
 - The type will be shown in the dialog box and should be
 - PKCS #12
 - PKCS #8
 - OpenSSL
 - c. Click OK
 - Import the certificate
 - a. Click Tools → Import Key Pair
 - b. Select the Key Pair Type of the certificate
 - c. Click OK
 - d. For PKCS#12 Certificate Types
 1. Enter the password of the certificate in the Decryption Password
 2. In the Key Pair File, click Browse and select the certificate to import
 3. Click Import
 4. The Alias can be left as is or enter a preferred name
 5. Click OK
 6. Enter the password, it will need to match the Keystore password
 7. Click OK and the certificate will be imported
 8. Save the Keystore file
 - e. For PKCS#8 or OpenSSL Certificate Types

2. If no Decryption Key needed, uncheck Encrypted Private Key
3. Click Browse next to the Key File, select the Key file and Click Choose
4. Click Browse next to the Certificate File, select the certificate file and Click Choose
5. Click Import
6. The Alias can be left as is or enter a preferred name
7. Click OK
8. Enter the password, it will need to match the Keystore password
9. Click OK and the certificate will be imported
10. Save the Keystore file

Creating a Truststore file

1. Open Keystore Explorer
2. Click on File → New
 - Select JCEKS or JKS for the Truststore Type
3. Click on File → Save
 - a. Enter a password for the truststore file
 - b. Click OK
 - c. Enter the name for the truststore file and click Save
4. Import the trusted certificate(s)
 - a. Click on Tools → Import Trusted Certificate

NOTE

For PFX files, you will need to extract the X.509 certificate using the steps to extract the X.509 certificate

- b. Select the certificate file to import
- c. Click Import
- d. Keep the Alias as is or enter a preferred name
- e. Save the Truststore file

Extract the X.509 certificate from the PFX file

1. Open the Keystore Explorer
2. Open the PFX file
 - a. Click File → Open
 - b. Select the PFX file

- c. Enter the password and click OK

3. Export the certificate
 - a. Double-click the certificate to open the details
 - b. If there is a hierarchy chain, click on the certificate authority to export
 - c. Click on Export
 - d. Set the Export Format to X.509
 - e. PEM checkbox can be left as is
 - f. Set the Export File to the name of the cer file
 - g. Click Export
4. Now you can import the certificate file to the truststore

Keystore and Truststore Generation with Java Keytool and OpenSSL

InstallerDialogBannerImage

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 - Message Broker server
 - Mattersight Enlighten Bridge server

Creating a Keystore file

1. Create a Keystore with the encryption key

- a. Open a command prompt
- b. Run the command:

```
keytool -genseckey -alias nexidia_encryption_key -keyalg AES -keysize 128 -keystore <keystore_filename>
```

- when prompted, enter a password for the keystore file
- press RETURN to use the same password for the nexidia_encryption_key

https://wem-dochub.mindtouch.us/Nexidia_Analytics_and_Data_Exchange_Framework/NXQC_12.5/Grid_Documentati...

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NOTE

All passwords used in the keystore file entries MUST be the same as the keystore password

2. Import a PFX certificate

- a. Get the alias from the PFX file, there should only be one certificate, the type will be PrivateKeyEntry
- b. Run the command:

```
keytool -keystore <pfx_file_to_import> -list
```

- when prompted, enter the <pfx_file_to_import> password
- The output should contain the following

```
Your keystore contains 1 entry
<alias_to_import>, Apr 5, 2024, PrivateKeyEntry,
Certificate fingerprint (SHA-256): <ignore_info>
```

- the <alias_to_import> will be the name of the certificate and used in the next steps

- a. Import the certificate from the <pfx_file_to_import> to the Keystore using the command

```
keytool -importkeystore -srckeystore <pfx_file_to_import> -srcstoretype pkcs12 -destkeystore <keystore_
filename>
```

- When prompted for the destination keystore password, enter the <keystore_filename> password assigned earlier
- When prompted for the source keystore password, enter the <pfx_file_to_import> password

- b. If needed, change the imported certificate in the Keystore to match the Keystore password with the command

```
keytool -keypasswd -keystore <keystore_filename> -alias <alias_to_import>
```

- Enter the <keystore_filename> password
- Enter the <pfx_file_to_import> password
- Set the new alias password to the Keystore password

1. Converting separate certificate and key files into a PFX file

- c. Run the command

```
openssl pkcs12 -export -in <certificate_file> -inkey <key_file> [-certfile <ca_cert_file>] -out <output_
pfx_file> -password pass:<keystore_password> -name <preferred_alias>
```

- where the option -certfile is only used if you have a Certificate Authority file available, usually this is with internally signed certificates. This will add a chain hierarchy to the PFX file

- d. Import the certificate from the <pfx_file_to_import> to the Keystore

Creating a Truststore file

1. Extract the root and intermediate CA certs from PFX file into a PEM file with the command:

```
openssl pkcs12 -in <pfx_file> -nokeys -cacerts -nodes -password pass:<pfx_password> -out <pem_output_file>
```

- a. If multiple certificates in the chain, then separate each CA cert into its own file, copy from -----BEGIN CERTIFICATE----- to -----END CERTIFICATE----- into its own PEM file, referenced as <ca_pem_filename>
- b. Determine the alias of each CA cert

```
keytool -printcert -file <ca_pem_filename>
```

- In the output look for the Owner and use the information in the CN section for the alias

Example Output

```
Owner: CN=DigiCert Global G2 TLS RSA SHA256 2020 CA1, O=DigiCert Inc, C=US
Issuer: CN=DigiCert Global Root G2, OU=www.digicert.com, O=DigiCert Inc, C=US
Serial number: cf5bd062b5602f47ab8502c23ccf066
Valid from: Mon Mar 29 20:00:00 EDT 2021 until: Sat Mar 29 19:59:59 EDT 2031
```

- The alias in this example would be **DigiCert Global G2 TLS RSA SHA256 2020 CA1**

2. Import the preferred CA certificates into the Truststore with the command:

```
keytool -importcert -keystore <truststore_to_use> -storepass <truststore_password> -file <created_pem_file> -noprompt -alias "<ca_alias_to_use>"
```

- This command will also create the Truststore file with the password in <truststore_password> if it does not exist. Use the command for each CA certificate needed.

Creating a PFX Using OpenSSL and a Self-Signed Certificate

- Create the client key and Certificate Signing Request (CSR)

```
Windows: set OPENSSL_CONF=path\to\openssl.cnf
Linux/Cygwin: OPENSSL_CONF=/path/to/openssl.cnf

openssl genrsa -out machine_cert_key_filename.key 4096

openssl req -new -key machine_cert_key_filename.key -out machine_cert_filename.csr
```

- Create extension.ext with the following content:

```
authorityKeyIdentifier=keyid,issuer
basicConstraints=CA:FALSE
keyUsage = digitalSignature, nonRepudiation, keyEncipherment, dataEncipherment
subjectAltName = @alt_names
```

```
[alt_names]
DNS.1 = <FQDN of machine for which certificate is being created>
IP.1 = <IP of machine for which certificate is being created>
```

- Ensure the CA, in this example grid-CA.crt, is imported.
- Create the machine certificate.

```
openssl x509 -req -in machine_cert_filename.csr -CA grid-CA.pem -CAkey grid-CA.key -CAserial grid-CA.srl -out machine_name.client.auth.crt -days 3650 -sha256 -passin pass:<password> -extfile extension.ext
```

- For the following, use the Grid Self-Signed Certificate Authority (CA) files available in <RTG repo>\doc\developer\smb.med.encrypt.cert\ca (or your own self-signed CA files if you choose to create new ones), and the key created in the first step above.
- Combine the files into a PFX

```
openssl pkcs12 -export -in machine_name.client.auth.crt -inkey machine_cert_key_filename.key -certfile grid-CA.crt -out machine_name.pfx -name <machine alias>
```